VEHICLE MANAGEMENT TAB

Add Vehicle

Test Case #	Functional Requirement, or Test Objective	Input data	Expected result/output
1	The system shall allow the text area in the Vehicle Management Tab to output “Missing input. Please try again.” if any of the inputs are missing.	● Leave the “Date Obtained” DatePicker blank. ● Test input(s): ○ Plate (Text Field): “58718D” ○ Date Obtained (Date Picker): null ○ Make (Radio Buttons): Ford ○ Campus (ComboBox): Busch, New Brunswick ○ Mileage (Text Field): 24390	“Missing input. Please try again.” in the output Text Area.
2	The system shall allow the text area in the Vehicle Management Tab to output “(wrong license plate) - license plate number must be exactly 6 characters.” if the license plate is not 6 tokens.	● The plate will be “5878D” which is only 5 tokens. ● Test input(s): ○ Plate (Text Field): “5878D” ○ Date Obtained (Date Picker): 10/22/2025 ○ Make (Radio Buttons): Ford ○ Campus (ComboBox): Busch, New Brunswick ○ Mileage (Text Field): 24390	“(wrong license plate) - license plate number must be exactly 6 characters.” in the output Text Area.
3	The system shall allow the text area in the Vehicle Management Tab to output “(date) - is today or a future date.” if the inputted date is a future date.	● The date will be “12/25/2025” in the DatePicker. ● Test input(s): ○ Plate (Text Field): “58718D” ○ Date Obtained (Date Picker): 12/25/2025 ○ Make (Radio Buttons): Ford ○ Campus (ComboBox): “Busch, New Brunswick” ○ Mileage (Text Field): 24390	“(date) - is today or a future date.” in the output Text Area.

4	The system shall allow the text area in the Vehicle Management Tab to output “(license plate) - last character is not a valid vehicle type.” if the inputted license plate doesnt end with a correct vehicle type character (must be capitalized).	● The license plate will be “32563F”. ● Test input(s): ○ Plate (Text Field): “32563F” ○ Date Obtained (Date Picker): 12/25/2025 ○ Make (Radio Buttons): Ford ○ Campus (ComboBox): “Busch, New Brunswick” ○ Mileage (Text Field): 24390	“(license plate) - last character is not a valid vehicle type.” in the output Text Area.
5	The system shall allow the text area in the Vehicle Management Tab to output the vehicle information string if all the inputs are correct and valid.	● All valid inputs for a vehicle. ● Test input(s): ○ Plate (Text Field): “58718D” ○ Date Obtained (Date Picker): 10/15/2025 ○ Make (Radio Buttons): Ford ○ Campus (ComboBox): “Busch, New Brunswick” ○ Mileage (Text Field): 24390	Vehicle String in the output Text Area.

Remove Vehicle

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall allow the text area in the Vehicle Management Tab to output “(license plate) is not in the fleet.” in the Text Area if the license plate is either invalid or not found in the fleet.	● The plate is “578D” ● Test input(s): ○ Plate (Text Field): “578D”	“(license plate) is not in the fleet.” in the output Text Area.
2	The system shall allow the text area in the Vehicle Management Tab to output “(license plate) has been removed from the fleet.” if the inputted license plate is valid and was found in the fleet.	● The plate is “58718D” (was in the fleet) ● Test input(s): ○ Plate (Text Field): “58718D”	“(license plate) has been removed from the fleet.” in the output Text Area.

VEHICLE BOOKING TAB

Loading Vehicles from a File

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system will output “Vehicle successfully loaded into the fleet” when a vehicle is successfully loaded from a text file into the fleet	• A text file with valid text: • Test input “58718D 2/29/2020 FORD 64390 Busch 58719D 1/31/2017 TOYOTA 87120 Livingston 34123X 3/29/2010 honda 51010 Camden 34448X 4/2/2023 FORD 19880 Livingston	The 4 vehicles would be added to the fleet, and “Vehicle successfully loaded into the fleet” in the output text area
2	The system will output “invalid line format” when there is not enough or too many arguments in a line of the text file.	• A text file with the invalid text • Test input “23 467 Abs “	The system will show “invalid line format” in the output text area
3	The system will output “Vehicle unsuccessfully loaded into the fleet” if there is enough arguments on the lines of the text files, but they are incorrect towards making a vehicle	• A text file with the invalid text: • Test Input “58718D 2/29/2020 FORD 64390 40820 58719D 1/31/2017 DDDD 87120 Livingston 34123X 3/29/2010 honda apple Camden 34448F 4/2/2023 FORD 19880 Livingston “	The system will show “Vehicle unsuccessfully loaded into the fleet” in the output text area

Cancel a reservation

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall output “Please fill out all fields before removing.” if any of the inputs are missing.	• Leave **Vehicle** blank. • Test input(s): Vehicle: *null* Start Date: 11/15/2025 End Date: 11/16/2025	The text area in the Vehicle reservation Tab will show “Please fill out all fields before removing.”
2	The system shall output “No	• Have the inputs not	“No matching

	matching booking found.” if the booking with the given details does not exist in the reservation list.	match any bookings currently in the reservation list ● Test input(s): Vehicle: 58719D Start Date: 11/15/2025 End Date: 11/16/2025	booking found.” in the output text area.
3	The system shall output “Booking removed: [booking details]” if the booking exists and is successfully canceled.	● Have the booking exactly match a booking in the reservation list ● Test input(s): Vehicle: 58718D (exists in reservation list) Start Date: 11/15/2025 End Date: 11/16/2025	“Booking removed: [Booking details string]” in the output text area.

Book a reservation

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall output “please fill out all fields before booking” if any of the booking inputs are missing.	● Leave the “Employee” ComboBox blank. ● Test inputs: Vehicle (ComboBox): 58718D Employee (ComboBox): null Campus (ComboBox): Busch Start Date (DatePicker): 11/22/2025 End Date (DatePicker): 11/25/2025	“Missing input. Please try again.” in the output text area.
2	The system shall output “Ending date cannot be before start date” if the start date is after the end date.	● Have a start date earlier than the end date ● Test inputs: Vehicle (ComboBox):58718D Employee (ComboBox): Lim Campus (ComboBox): Busch Start Date: 11/28/2025 End Date: 11/22/2025	“Ending date cannot be before starting date.” in the output text area.
3	The system shall output “Cannot book a duration more than a week” if the difference between start and end date is greater than 7 days.	● Create a booking that exceeds 7 days ● Test inputs: Vehicle: 58718D Employee: Lim Campus: Busch Start Date: 11/01/2025 End Date: 11/12/2025	“11/01/2025 ~ 11/12/2025 - cannot book a duration more than a week.” in the output text area.

4	The system shall output “Cannot book more than 3 months in advance” if the start date is beyond 3 months in the future.	• Create a booking that starts more than 3 months from the current day • Test inputs: Vehicle:58718D Employee: Lim Campus: Busch Start Date: 02/15/2026 End Date: 02/20/2026	“Cannot book more than 3 months in advance.” in the output text area.
5	The system shall output “Vehicle is not available on the selected dates.” if the chosen vehicle is already booked during the selected date range.	• Create a booking where the vehicle is already booked on the days you are trying to book it for • Test inputs: Vehicle: 58718D (already booked 10/28–11/02) Employee: Lim Campus: Busch Start Date: 10/29/2025 End Date: 10/31/2025	“Vehicle 58718D is not available on those dates.” in the output text area.
6	The system shall output “Cannot book a vehicle in the past” if the starting date is before the current date	• Create a booking where the current date is after the start date of the booking • Test inputs: Vehicle: 58718D Employee:Lim Campus: Livingston Start Date: 10/22/2025 End Date: 10/23/2025	“Cannot book a vehicle in the past.” in the output text area.
7	The system shall output “Employee already has a booking that overlaps these dates.” if the employee has another reservation that overlaps the entered dates.	• Create a booking for an employee on dates they are already booked for. • Test inputs: Vehicle: 58718D Employee: Lim (already has 10/29–10/31 booking) Campus: Busch Start Date: 10/30/2025 End Date: 11/02/2025	“Employee Lim already has a booking that overlaps these dates.” in the output text area.
8	The system shall output “Booking Confirmed.” when all inputs are valid and	• Create a valid booking • Test inputs: Vehicle: 58718D	“Booking confirmed: [Booking details string]” in the output

	constraints are met.	Employee: Lim Campus: Busch Start Date: 10/29/2025 End Date: 10/31/2025	text area.

VEHICLE RETURNING TAB

Completing Trip

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall output “Please select a booking to complete. If no vehicles are in the booking list, please go book a vehicle in the Vehicle Reservation tab.” if no booking is selected.	• Don't select a booking • Test inputs Booking: *null* Mileage: 50000	“Please select a booking to complete. If no vehicles are in the booking list, please go book a vehicle in the Vehicle Reservation tab.” in the output area
2	The system shall output “Please enter the new mileage.” if the mileage field is empty.	• Don't enter a milage number • Test inputs Booking: Valid booking with license plate(58718D) Mileage: *blank*	“Please enter the new mileage.” in the output area
3	The system shall output “58718D booked with end date 11/02/2025 - returning not in order of ending date.” if the booking is not the earliest one to end.	• Select a booking that ends later than a booking with the same car that ends earlier • Test inputs Booking: 58718D (end date 11/02/2025, but earlier booking exists) Mileage: 55000	“58718D booked with end date 11/02/2025 - returning not in order of ending date.” in the output area
4	The system shall output “Mileage must be a numeric value.” if the entered mileage is not numeric.	• Input a non-numeric value for mileage • Test inputs Booking: Valid booking (58718D) Mileage: fifty thousand	“Mileage must be a numeric value.” in the output area
5	The system shall output	• Input 0 or less for	“Invalid mileage

	“Invalid mileage (cannot be less than or equal to 0).” if the mileage entered is zero or negative.	mileage ● Test inputs Booking: Valid booking (58718D) Mileage: 0	(cannot be less than or equal to 0).” in the output area
6	The system shall output “New mileage must be greater than old mileage.” if the new mileage is not greater than the old mileage.	● Input a mileage < the car’s current milage ● Test inputs Booking: Valid booking (58718D) Old mileage of car: 50000 New mileage: 49000	“New mileage must be greater than old mileage.” in the output area
7	The system shall output “Trip completed: [Trip details]” if the trip is completed successfully.	● Input a valid mileage and select a valid booking ● Test inputs Booking: Valid booking (58718D) mileage: 50000	“Trip completed: [Trip details]” in the output area

VEHICLE REPORTING TAB

Printing Reservations

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall allow the text area in the Vehicle Reporting Tab to output “There is no booking record.” if there are no bookings in the reservation list in any dropdown option.	● Empty reservation list.	“There is no booking record.” outputted in the text area.
2	The system shall allow the text area in the Vehicle Reporting Tab to output the reservations listed in the order mentioned by the dropdown if there are bookings in the reservation list.	● Non-empty reservation list. ● Whichever dropdown option was picked.	Bookings listed in the order mentioned by the dropdown in the output Text Area.

Printing Fleet

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall allow the text area in the Vehicle Reporting Tab to output “There is no booking record.” if there are no bookings in the reservation list.	● Empty fleet list.	“There is no vehicle in the fleet.” outputted in the text area.
2	The system shall allow the text area in the Vehicle Reporting Tab to output the fleets listed ordered by location/make/date.	● Non-empty fleet list.	Vehicles listed in the order mentioned outputted in the text area.

Printing CompletedTrips/Cost Report

Test Case #	Functional Requirement, or Test Objective	Test description and input data	Expected result/output
1	The system shall allow the text area in the Vehicle Reporting Tab to output “There is no completed trips.” in the Text Area if there are no trips in the triplist in the “by plate/ending date (PT)” option.	● Empty triplist list.	“There is no completed trips.” outputted in the text area.
2	The system shall allow the text area in the Vehicle Reporting Tab to output “There is no archived trips for the cost report.” in the Text Area if there are no trips in the triplist in the “cost report by department (PC)” option.	● Empty triplist list.	“There is no archived trips for the cost report.” outputted in the text area.
3	The system shall allow the text area in the Vehicle Reporting Tab to output the trips listed in the order of plate then ending date if the dropdown option that was chosen was “by plate/ending date (PT)”	● Non-empty triplist list. ● Dropdown option was “by plate/ending date (PT)”	Trips listed in the order of plate then ending date outputted in the text area.
4	The system shall allow the text area in the Vehicle Reporting Tab to output the cost report of	● Non-empty triplist list. ● Dropdown option was “cost report by department (PC)”	Cost report trips ordered by department outputted

	the trips ordered by department if the dropdown option that was chosen was “cost report by department (PC)”.		in the text area.